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SCHOOL OF COMPUTER SCIENCE ENGINEERING

COURSE: INT 387

Data Analysis And Visualization

SUBMITTED TO: **Dr. Savleen Kaur**

TOPIC

“GST Revenue Insights Dashboard ”

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Under the Guidance of
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1. INTRODUCTION:

The Goods and Services Tax (GST) is one of the most important sources of revenue for the Government of India. Analyzing GST data helps in understanding the economic performance of different states and identifying revenue trends.

In this project, an interactive dashboard is developed using Microsoft Power BI to analyze GST revenue data. The dashboard provides insights into revenue distribution, tax components, regional performance, and monthly trends.

The aim of this project is to transform raw data into meaningful insights using data visualization techniques.

2. SOURCE OF DATASET

The dataset used in this project is obtained from the National Data and Analytics Platform (NDAP), which provides authentic and reliable government data.

The dataset contains:

- State-wise GST revenue
- Tax components (CGST, SGST, IGST, Cess)
- Monthly and yearly data

Screenshot of NDAP website:

→ ↻ ndap.niti.gov.in/dataset/7114 ☆ 📄 🔍

New Tab All Bookmarks

 Newsletter Data Catalogue → Screen Reader English A+ A A- Help

 Tax Revenue Collection on Goods & Service Tax (GST) Portal

Standardised By NDAP

Profile Data Table Visualisation Share Subscribe Save Download Table

Overview

Under the Goods and Services Tax (GST) law, the Central Government will collect Central Goods and Services Tax (CGST), State Goods and Services Tax (SGST), Integrated Goods and Services Tax (IGST) or CESS, depending upon whether the transaction is intrastate or interstate. When the supply of goods or services happens within a state, called intra-state transactions, then both the Central Goods and Services Tax (CGST) and State Goods and Services Tax (SGST) will be collected. Whereas if the supply of goods or services happens between the states and inter-state transactions, then only Integrated Goods and Services Tax (IGST) will be collected. And Cess is a form of tax charged/levied over and above the base tax liability of a taxpayer.

Source :
Ministry of Finance

[More Info](#) • [Jump to Metadata](#)

 Data published by | Ministry/ Department/ Organisation

Ministry of Finance

 Collections

[Goods and Services Tax \(GST\) Statistics](#)

 Geographical Coverage

State

 Frequency

Yearly

 Sector

Finance, Banking & Insurance

 Dataset hosted by

NDAP

 Time Granularity

Monthly

 Year Range

FY 2017 - FY 2025

 Date Updated on

Aug 11, 2025

[View Processing Report](#)

Data Preview

3456 Rows | Indicators 5/5

Dataset Link:

https://ndap.niti.gov.in/dataset/7114?filter_id=5294

Website Link:

<https://ndap.niti.gov.in/>

3. DATASET PREPROCESSING

The dataset was preprocessed using Power BI Power Query Editor to ensure data quality and consistency.

The following steps were performed:

- Removed missing/null values
- Converted data types
- Created Month and Year columns
- Sorted month names correctly
- Calculated Total Revenue
- Cleaned and structured data

Raw dataset view:

1.2 Central Goods And Services Tax (Cgst) Revenue (UOM:INR{In...)	1.2 State Goods And Services Tax (Sgst)Revenue (UOM:INR{India...	1.2 Integrated Goods And Services Tax (Igst)Revenue (UOM:INI
978 distinct, 957 unique	985 distinct, 970 unique	981 distinct, 966 unique
12.59	15.28	
924.15	1225.4	
33.88	46.71	
392.15	522.15	
447.92	842.07	
43.97	58.62	
640.33	746.47	
1135.27	1315.91	
153.78	193.12	
3141.5	3790.78	
1638.59	2119.55	
153.23	213.67	
156.78	214.19	
610.29	781.24	
3182.2	3854.8	
1055.46	1304.6	
11.07	13.76	
0.5	0.49	
927.69	1167.62	
8477.11	9839.21	
13.93	18.79	
28.06	35.73	
9.95	14.7	
32.51	41.91	
1198.32	1506.2	
31.38	47.54	
539.68	782.67	
1125.74	1514.77	
22.66	30.56	
2886.21	3727.98	

Table.TransformColumnTypes(#"Promoted Headers",{"Country", type text}, {"State", type text}, {"Year", type text}, {"Month", type date},

	Country	State	Year	Month	1.2 Central Goods And Services Tax (Cgst) Revenue (U
	1 distinct, 0 unique	36 distinct, 0 unique	4 distinct, 0 unique	28 distinct, 0 unique	978 distinct, 957 unique
1	India	Andaman and Nicobar Islands	Financial Year (Apr - Mar), 2025	01-06-2025	
2	India	Andhra Pradesh	Financial Year (Apr - Mar), 2025	01-06-2025	
3	India	Arunachal Pradesh	Financial Year (Apr - Mar), 2025	01-06-2025	
4	India	Assam	Financial Year (Apr - Mar), 2025	01-06-2025	
5	India	Bihar	Financial Year (Apr - Mar), 2025	01-06-2025	
6	India	Chandigarh	Financial Year (Apr - Mar), 2025	01-06-2025	
7	India	Chhattisgarh	Financial Year (Apr - Mar), 2025	01-06-2025	
8	India	Delhi	Financial Year (Apr - Mar), 2025	01-06-2025	
9	India	Goa	Financial Year (Apr - Mar), 2025	01-06-2025	
10	India	Gujarat	Financial Year (Apr - Mar), 2025	01-06-2025	
11	India	Haryana	Financial Year (Apr - Mar), 2025	01-06-2025	
12	India	Himachal Pradesh	Financial Year (Apr - Mar), 2025	01-06-2025	
13	India	Jammu and Kashmir	Financial Year (Apr - Mar), 2025	01-06-2025	
14	India	Jharkhand	Financial Year (Apr - Mar), 2025	01-06-2025	
15	India	Karnataka	Financial Year (Apr - Mar), 2025	01-06-2025	
16	India	Kerala	Financial Year (Apr - Mar), 2025	01-06-2025	
17	India	Ladakh	Financial Year (Apr - Mar), 2025	01-06-2025	
18	India	Lakshadweep	Financial Year (Apr - Mar), 2025	01-06-2025	
19	India	Madhya Pradesh	Financial Year (Apr - Mar), 2025	01-06-2025	
20	India	Maharashtra	Financial Year (Apr - Mar), 2025	01-06-2025	
21	India	Manipur	Financial Year (Apr - Mar), 2025	01-06-2025	
22	India	Meghalaya	Financial Year (Apr - Mar), 2025	01-06-2025	
23	India	Mizoram	Financial Year (Apr - Mar), 2025	01-06-2025	
24	India	Nagaland	Financial Year (Apr - Mar), 2025	01-06-2025	
25	India	Odisha	Financial Year (Apr - Mar), 2025	01-06-2025	
26	India	Puducherry	Financial Year (Apr - Mar), 2025	01-06-2025	
27	India	Punjab	Financial Year (Apr - Mar), 2025	01-06-2025	
28	India	Rajasthan	Financial Year (Apr - Mar), 2025	01-06-2025	
29	India	Sikkim	Financial Year (Apr - Mar), 2025	01-06-2025	

Power Query Editor

Cleaning steps:

Table.TransformColumnTypes(#"Promoted Headers",{"Country", type text}, {"State", type text}, {"Year", type text}, {"Month", type date},

	Month	1.2 Central Goods And Services Tax (Cgst) Revenue (UOM:INR(In...	1.2 State Goods And
	28 distinct, 0 unique	978 distinct, 957 unique	985 distinct, 970 unique
I Year (Apr - Mar), 2025	01-06-2025	2.59	
I Year (Apr - Mar), 2025	01-06-2025	4.15	
I Year (Apr - Mar), 2025	01-06-2025	3.88	
I Year (Apr - Mar), 2025	01-06-2025	2.15	
I Year (Apr - Mar), 2025	01-06-2025	7.92	
I Year (Apr - Mar), 2025	01-06-2025	3.97	
I Year (Apr - Mar), 2025	01-06-2025	0.33	
I Year (Apr - Mar), 2025	01-06-2025	5.27	
I Year (Apr - Mar), 2025	01-06-2025	3.78	
I Year (Apr - Mar), 2025	01-06-2025	41.5	
I Year (Apr - Mar), 2025	01-06-2025	8.59	
I Year (Apr - Mar), 2025	01-06-2025	3.23	
I Year (Apr - Mar), 2025	01-06-2025	6.78	
I Year (Apr - Mar), 2025	01-06-2025	0.29	
I Year (Apr - Mar), 2025	01-06-2025	82.2	
I Year (Apr - Mar), 2025	01-06-2025	5.46	
I Year (Apr - Mar), 2025	01-06-2025	1.07	
I Year (Apr - Mar), 2025	01-06-2025	0.5	
I Year (Apr - Mar), 2025	01-06-2025	927.69	
I Year (Apr - Mar), 2025	01-06-2025	8477.11	
I Year (Apr - Mar), 2025	01-06-2025	13.93	
I Year (Apr - Mar), 2025	01-06-2025	28.06	
I Year (Apr - Mar), 2025	01-06-2025	9.95	
I Year (Apr - Mar), 2025	01-06-2025	32.51	
I Year (Apr - Mar), 2025	01-06-2025	1198.32	
I Year (Apr - Mar), 2025	01-06-2025	31.38	
I Year (Apr - Mar), 2025	01-06-2025	539.68	
I Year (Apr - Mar), 2025	01-06-2025	1136.74	

Country

State

Year

Month

1.2

CGST Revenue

1.2

SGST Revenue

1 distinct, 0 unique	36 distinct, 0 unique	4 distinct, 0 unique	28 distinct, 0 unique	978 distinct, 957 unique	985 distinct, 970 unique
India	Andaman and Nicobar Islands	Financial Year (Apr - Mar), 2025	01-06-2025	12.59	
India	Andhra Pradesh	Financial Year (Apr - Mar), 2025	01-06-2025	924.15	
India	Arunachal Pradesh	Financial Year (Apr - Mar), 2025	01-06-2025	33.88	
India	Assam	Financial Year (Apr - Mar), 2025	01-06-2025	392.15	
India	Bihar	Financial Year (Apr - Mar), 2025	01-06-2025	447.92	
India	Chandigarh			43.97	
India	Chhattisgarh			40.33	
India	Delhi			35.27	
India	Goa			53.78	
India	Gujarat			141.5	
India	Haryana			38.59	
India	Himachal Pradesh			53.23	
India	Jammu and Kashmir			56.78	
India	Jharkhand			10.29	
India	Karnataka			182.2	
India	Kerala			55.46	
India	Ladakh			11.07	
India	Lakshadweep	Financial Year (Apr - Mar), 2025	01-06-2025	0.5	
India	Madhya Pradesh	Financial Year (Apr - Mar), 2025	01-06-2025	927.69	
India	Maharashtra	Financial Year (Apr - Mar), 2025	01-06-2025	8477.11	
India	Manipur	Financial Year (Apr - Mar), 2025	01-06-2025	13.93	
India	Meghalaya	Financial Year (Apr - Mar), 2025	01-06-2025	28.06	
India	Mizoram	Financial Year (Apr - Mar), 2025	01-06-2025	9.95	
India	Nagaland	Financial Year (Apr - Mar), 2025	01-06-2025	32.51	

Text Before Delimiter

Enter the delimiter that marks the end of what you would like to extract.

Delimiter

Advanced options

OK

Cancel

Country

State

Year

Month

1.2

CGST Revenue

1.2

SGST Revenue

1 distinct, 0 unique	36 distinct, 0 unique	4 distinct, 0 unique	28 distinct, 0 unique	978 distinct, 957 unique	985 distinct, 970 unique
India	Andaman and Nicobar Islands	Financial Year (Apr - Mar), 2025	01-06-2025	12.59	
India	Andhra Pradesh	Financial Year (Apr - Mar), 2025	01-06-2025	924.15	
India	Arunachal Pradesh	Financial Year (Apr - Mar), 2025	01-06-2025	33.88	
India	Assam	Financial Year (Apr - Mar), 2025	01-06-2025	392.15	
India	Bihar	Financial Year (Apr - Mar), 2025	01-06-2025	447.92	
India	Chandigarh			43.97	
India	Chhattisgarh			40.33	
India	Delhi			35.27	
India	Goa			53.78	
India	Gujarat			141.5	
India	Haryana			38.59	
India	Himachal Pradesh			53.23	
India	Jammu and Kashmir			56.78	
India	Jharkhand			10.29	
India	Karnataka			182.2	
India	Kerala			55.46	
India	Ladakh			11.07	
India	Lakshadweep	Financial Year (Apr - Mar), 2025	01-06-2025	0.5	
India	Madhya Pradesh	Financial Year (Apr - Mar), 2025	01-06-2025	927.69	
India	Maharashtra	Financial Year (Apr - Mar), 2025	01-06-2025	8477.11	
India	Manipur	Financial Year (Apr - Mar), 2025	01-06-2025	13.93	
India	Meghalaya	Financial Year (Apr - Mar), 2025	01-06-2025	28.06	
India	Mizoram	Financial Year (Apr - Mar), 2025	01-06-2025	9.95	
India	Nagaland	Financial Year (Apr - Mar), 2025	01-06-2025	32.51	

Text Before Delimiter

Enter the delimiter that marks the end of what you would like to extract.

Delimiter

Advanced options

OK

Cancel

After Cleaning:

A ^B _C Country	A ^B _C State	A ^B _C Year	Month	1.2 CGST Revenue	1.2 SGST Revenue
1 distinct, 0 unique	36 distinct, 0 unique	4 distinct, 0 unique	28 distinct, 0 unique	978 distinct, 957 unique	985 distinct, 970
India	Andaman and Nicobar Islands	2025	01-06-2025	12.59	
India	Andhra Pradesh	2025	01-06-2025	924.15	
India	Arunachal Pradesh	2025	01-06-2025	33.88	
India	Assam	2025	01-06-2025	392.15	
India	Bihar	2025	01-06-2025	447.92	
India	Chandigarh	2025	01-06-2025	43.97	
India	Chhattisgarh	2025	01-06-2025	640.33	
India	Delhi	2025	01-06-2025	1135.27	
India	Goa	2025	01-06-2025	153.78	
India	Gujarat	2025	01-06-2025	3141.5	
India	Haryana	2025	01-06-2025	1638.59	
India	Himachal Pradesh	2025	01-06-2025	153.23	
India	Jammu and Kashmir	2025	01-06-2025	156.78	
India	Jharkhand	2025	01-06-2025	610.29	
India	Karnataka	2025	01-06-2025	3182.2	
India	Kerala	2025	01-06-2025	1055.46	
India	Ladakh	2025	01-06-2025	11.07	
India	Lakshadweep	2025	01-06-2025	0.5	
India	Madhya Pradesh	2025	01-06-2025	927.69	
India	Maharashtra	2025	01-06-2025	8477.11	
India	Manipur	2025	01-06-2025	13.93	
India	Meghalaya	2025	01-06-2025	28.06	
India	Mizoram	2025	01-06-2025	9.95	
India	Nagaland	2025	01-06-2025	32.51	
India	Odisha	2025	01-06-2025	1198.32	
India	Puducherry	2025	01-06-2025	31.38	
India	Punjab	2025	01-06-2025	539.68	
India	Rajasthan	2025	01-06-2025	1125.74	
India	Sikkim	2025	01-06-2025	22.66	
India	Tamil Nadu	2025	01-06-2025	2886.21	

	Month	1.2 CGST Revenue	1.2 SGST Revenue	1.2 IGST Revenue	1.2 Cess Revenue	1.2 Total Revenue
	28 distinct, 0 unique	978 distinct, 957 unique	985 distinct, 970 unique	981 distinct, 966 unique	745 distinct, 673 unique	998 distinct, 996 unique
, 2025	01-06-2025	12.59	15.28	4.12	0.03	32.02
, 2025	01-06-2025	924.15	1225.4	1228.59	256.23	3634.37
, 2025	01-06-2025	33.88	46.71	7.46	0.16	88.21
, 2025	01-06-2025	392.15	522.15	358.18	132.04	1404.52
, 2025	01-06-2025	447.92	842.07	238.43	180.48	1708.9
, 2025	01-06-2025	43.97	58.62	111.96	5.04	219.59
, 2025	01-06-2025	640.33	746.47	1204.06	685.2	3276.06
, 2025	01-06-2025	1135.27	1315.91	3075.33	83.29	5609.8
, 2025	01-06-2025	153.78	193.12	199.84	4.25	550.99
, 2025	01-06-2025	3141.5	3790.78	3558.62	549.22	11040.12
, 2025	01-06-2025	1638.59	2119.55	5478.42	722.69	9959.25
, 2025	01-06-2025	153.23	213.67	527.82	7.73	902.45
, 2025	01-06-2025	156.78	214.19	181.14	13.85	565.96
, 2025	01-06-2025	610.29	781.24	1036.73	658.09	3086.35
, 2025	01-06-2025	3182.2	3854.8	4747.49	1624.07	13408.56

Name

GST_Revenue_Dashboard

All Properties

APPLIED STEPS

Source	⚙
✕ Promoted Headers	⚙
Changed Type	
Renamed Columns	
Extracted Text After Delimiter	⚙
Inserted Month Name	⚙
Trimmed Text	
Inserted Month	⚙
Removed Blank Rows	
Removed Blank Rows1	
Inserted Year	⚙
Removed Columns	
Inserted Year1	⚙
Removed Columns1	
Reordered Columns	
Renamed Columns1	
Changed Type1	

4. ANALYSIS ON DATASET

☐ i. General Description

The dataset represents GST revenue collected across various states over time. It includes multiple tax components and provides a comprehensive view of revenue distribution.

☐ ii. Specific Requirements

The objectives of this project are:

- Analyze GST revenue
- Identify top and lowest performing states
- Study tax component contribution
- Analyze monthly trends
- Build an interactive dashboard

☐ iii. Analysis Results

The analysis revealed the following insights:

- Maharashtra is the highest revenue-generating state
- Some states contribute very low revenue
- GST revenue varies across months
- SGST and CGST contribute the highest share

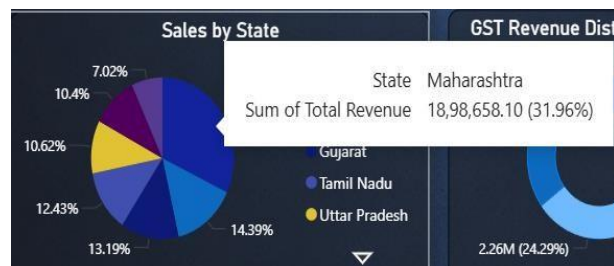
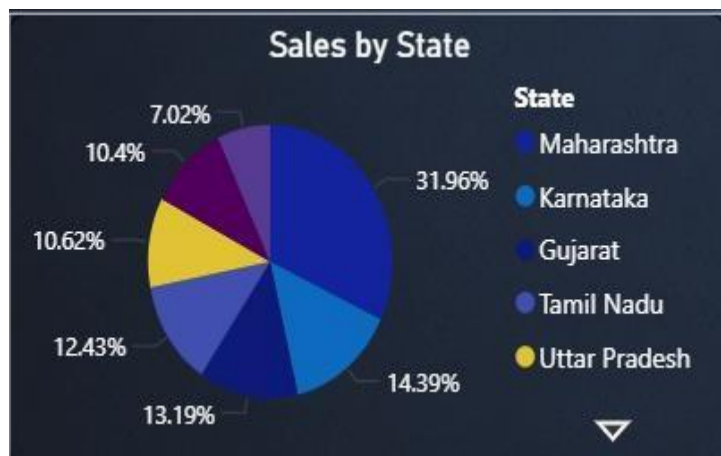
☒ iv. Visualization

The insights are visualized using Power BI dashboard with various charts.

✓ 1. FULL DASHBOARD



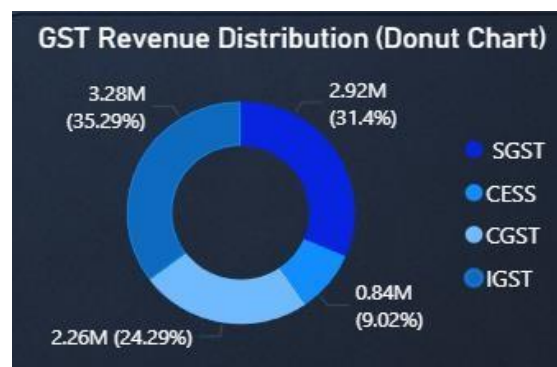
✓ 2. PIE CHART



State-wise Revenue Distribution

This chart shows the contribution of each state to total GST revenue.

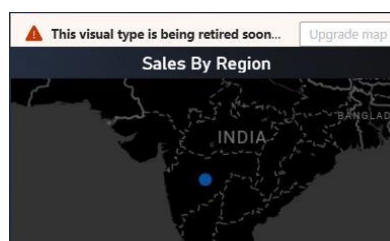
✓ 3. DONUT CHART



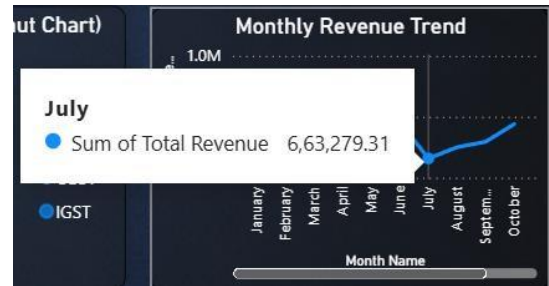
GST Component Distribution

This chart represents CGST, SGST, IGST, and Cess contribution.

✓ 4. MAP (State-wise Revenue Map)



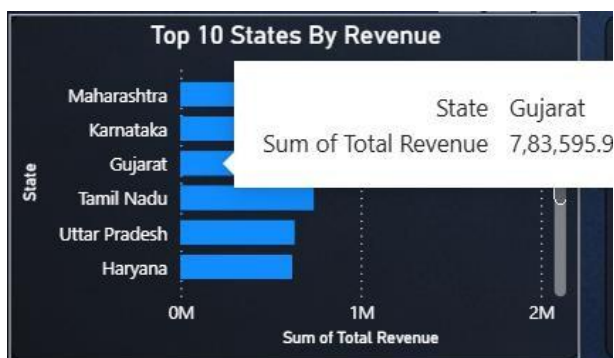
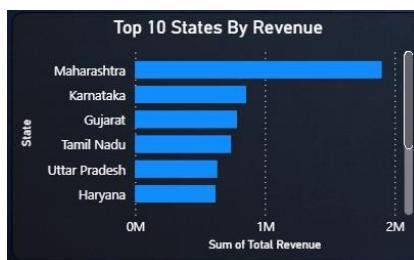
✓ 5. LINE CHART



Monthly Revenue Trend

This chart shows variation of GST revenue over time.

✓ 6. BAR CHART



Funnel Chart



Top and Lowest Performing State

5. CONCLUSION

The project successfully demonstrates how data visualization can be used to analyze GST revenue. The dashboard provides clear insights into state performance, tax distribution, and trends. It helps in understanding which states contribute more and how revenue changes over time.

6. FUTURE SCOPE

- Integration of real-time data
- Use of machine learning for prediction
- Advanced filtering and drill-through
- Enhanced map visualization

7. REFERENCES

1. NDAP Government Portal – <https://ndap.niti.gov.in/>
2. Microsoft Power BI Documentation
3. Online learning resources

